

AVNISH RAJ

Software Engineer | Full-Stack & AI Systems

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SUMMARY

Fifth-semester B.Tech Computer Science student building production-minded AI systems, developer tools, and full-stack applications. Experienced with TypeScript, Python, Next.js, React, Node.js, databases, REST APIs, cloud integrations, automated testing, and secure application design.

EDUCATION

Thapar Institute of Engineering and Technology

2024 - 2028

B.Tech Computer Science | CGPA: 7.71/10

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, Solidity

Frameworks & Libraries: React, Next.js, Node.js, Express, Vite, Tailwind CSS

Databases & Platforms: MySQL, Firebase, Cloud Firestore, Supabase, Vercel, Base Sepolia

Tools & Concepts: Git, GitHub, REST APIs, AI/LLM Integration, PWA, Automated Testing, DSA, DBMS, OOP, Operating Systems

PROJECTS

SCAR: Production's Immune System | GitHub | Live Demo

Next.js, TypeScript, React, Hindsight Cloud, Groq, Zod, Vitest

- Built a deployment-risk agent that retains production incidents, recalls causally similar failures, and changes future deployment verdicts using persistent organizational memory.
- Implemented server-side API routes, structured AI risk analysis, evidence-citation gates, isolated demo sessions, deterministic fallback logic, and security controls; verified with 17 automated tests.

Lifeline | GitHub

Python, CLI, Unix PTY, JSONL, Security

- Developed a dependency-free Python CLI that captures interrupted AI coding sessions and resumes work across Claude Code, Codex, and Gemini without manual context reconstruction.
- Engineered cross-tool session readers, automatic usage-limit detection, secret redaction, restrictive file permissions, path validation, and argv-based process launching to reduce credential and command-injection risk.

Campus Lost & Found PWA | GitHub | Live Demo

JavaScript, Firebase Authentication, Cloud Firestore, Cloudinary, PWA

- Created and deployed a campus-exclusive Progressive Web App for reporting and recovering lost items with intelligent matching, real-time notifications, search, filtering, and moderation.
- Integrated restricted Google authentication, Firestore persistence, Cloudinary image uploads with client-side compression, auto-expiring posts, and an installable service-worker experience.

UGF Skill Passport | GitHub | Contract

React, TypeScript, Solidity, ethers.js, Base Sepolia, Vite

- Built a Web3 quest application that mints onchain NFT skill badges through a deployed Solidity contract while allowing users to claim without holding ETH for gas.
- Integrated wallet connection, Base Sepolia network handling, UGF gas abstraction, alternate-token fee settlement, and duplicate-claim prevention with verifiable onchain proof.